

Cocycle Transport and Ordered Boundary Interfaces for Unknot Recognition

Exact Pachner scores, checked geometric descent,
and four-coordinate recovery of compressed attachment order

Research continuation for the ProveIt project

9 October 2026

Abstract

We develop two concrete improvements to the geometric part of an unknot recognition program. First, an integral simplicial cocycle is transported through a checked 2–3 or 3–2 Pachner move using five integer heights. We derive exact, constant-size formulas for the Euler-characteristic change and the change in the number of normal discs of the coherent fibre. They permit all eligible collapses to be scored in one incidence scan and remove repeated cohomology elimination from a downward epoch. The edge max-norm does not increase under 3–2 moves and grows by at most a factor of two under each 2–3 move, giving an explicit bound on every intermediate binary coordinate.

Second, we recover the cyclic order of finitely many marked occurrences on a binary-encoded normal boundary curve. Removing the marks leaves paths and unmarked circles. Because every path has exactly two endpoint occurrences, its endpoint pair is recovered exactly from a count, a sum, and a sum of squares. Together with path length, four additive coordinates replace a vector whose dimension grows with the number of marks. This is a deterministic encoding, with independently checked decoding and no probabilistic fingerprints.

The implementation includes separate certificate consumers, source-bound unknot witnesses, literal and Regina comparisons, adversarial tests, and paired performance experiments. The local results advance retriangulation and attachment handling. They do not bound the number of geometric escapes needed for arbitrary diagrams, nor supply the complete three-dimensional cutting compiler required by a general quasi-polynomial recognizer.

Audited repository revision

[3a90fb34146c915328ab8eac6250cc2514f74ed0](#)

Main target: `Topology/UnknotRecognition/fast/`.

The supplied patch is additive; the existing recognition schedule remains unchanged.

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1 Results, scope, and the current baseline

The useful next question is how to carry a compressed geometric object through an actual change of triangulation, and how to retain the order information needed to attach its boundary. An inventory of components is insufficient for these tasks. Likewise, a fast certificate checker does not give a bound for discovering a successful certificate. The present work addresses two of these concrete interfaces and makes the remaining search assumption explicit.

1.1 What has been established

The mathematical and implemented contributions are as follows.

1. An exact formula for the coherent-fibre Euler jump across a formal bipyramid, valid for arbitrarily large integral heights and for the generalized triangulations admitted by the checked move interface.
2. An exact nonnegative formula for the corresponding normal-disc increase in the 2–3 direction. Together, the formulas supply a deterministic score for 3–2 candidate discovery.
3. A cochain-preserving implementation of a complete downward epoch, with independent replay of every move and an explicit coordinate-bit bound. It solves cohomology once at entry, not once per accepted collapse.
4. A source-bound positive-certificate consumer. It checks the input diagram’s compact exterior, any preliminary boundary shellings, the transported sequence, and a terminal essential-disc component certificate.
5. Exact recovery of marked boundary cycles, occurrence identities, intervening lengths, and unmarked-cycle multiplicities. A four-coordinate moment representation removes the increasing weight dimension from the endpoint-identification query.

The algebraic formulas and the two-endpoint decoding theorem are proved below. The orbit engine, manifold validator, and normal-component kernels are pre-existing project components. Their use is stated explicitly rather than counted as newly implemented functionality.

1.2 What the audited repository already contains

The audit is pinned to `3a90fb34146c915328ab8eac6250cc2514f74ed0`; the live repository may continue to change. At this revision, the main code already contains the direct diagram-to-exterior construction and its independent replay, boundary shellings, checked Pachner moves in both directions, primitive rank-one cocycle extraction, minimum-span optimization, exact Euler optimization on a certified minimum-span face, and compressed normal-component queries. In particular, the diagram-to-exterior correspondence is no longer an unimplemented entry step at this baseline [7].

The earlier coherent-escape study supplies an eight-tetrahedron solid-torus example whose primitive coherent surface has genus two. It exhibits a 3–2 escape and investigates greedy simplification followed by fresh cocycle extraction. The present contribution continues that experiment with direct cochain transport and exact candidate scores. The old study does not contain the five-height jump formulas or the moment endpoint encoding derived here.

The incoming inventory was also checked. Ten relevant packages cover quadrilateral-support kernels, sparse normal and port incidence, topology spectra, compiled orbit certificates, weighted component certificates, power-conjugacy methods, and selective orbit transfer. Their component and profile interfaces are useful predecessors, but the inspected port packages explicitly leave cyclic order and arbitrary pointwise attachment maps outside their contracts. The present ordering interface addresses finitely many distinct marked occurrences; it does not silently extend to every pointwise gluing operation. The package contains a separate source audit identifying these boundaries.

1.3 Relation to primary literature

Agol–Hass–Thurston supply the polynomial orbit and weighted-orbit machinery for interval pairings [1]. Lackenby’s efficient-certification paper uses such computations to recover ordered intersections around vertical parallelity annuli; its proof of Theorem 9.2, especially Section 9.4, is the theoretical predecessor of the marked-order interface [3]. Our implementation specializes that established method and reduces the number of additive coordinates by exploiting the two-endpoint property.

Normal surfaces under Pachner moves, including the annulus-to-two-discs configuration, are studied by Garoufalidis–Hodgson–Hoffman–Rubinstein [2, Section 12.4]. Their discussion makes an essential distinction: an arbitrary normal surface may be transported with its topology preserved, whereas transporting a cocycle and choosing its coherent fibre can change the surface. Our exact score is for the latter operation. We claim a derived formula and checked implementation in this project, not priority for the underlying local surgery picture.

Polynomial-bit transport of cohomology during boundary simplification is also established in Lackenby’s Proposition 10.3 [3]. The elementary max-norm estimate below is specialized to individual checked 2–3 and 3–2 moves.

Lackenby’s 2021 slides state the announced running time $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ [4]. His July 2026 preprint develops hierarchy algorithms and efficient certificates, including the compressed cutting interface of Theorem 14.4, but does not state a general quasi-polynomial runtime theorem [5]. We therefore use *quasi-polynomial* in its usual broad sense $\exp((\log n)^{O(1)})$, and state separately when a conditional result gives the stronger $n^{O(\log n)}$ form.

2 Integral cocycles and coherent normal fibres

2.1 Encoding conventions

Let $\mathcal{T}$ triangulate a compact three-manifold with t tetrahedra. Face pairings are explicitly listed, while normal coordinates and cocycle values are binary integers. Formal tetrahedron vertices may become identified in the manifold. An oriented edge cocycle z assigns an integer to every oriented global edge, reverses sign under reversal, and has zero oriented sum around every triangular face.

The native adapter further requires a connected orientable manifold with exactly one torus boundary component. Closed, ideal, disconnected, and other-boundary triangulations are outside this adapter’s present contract. The local identities below do not need all these additional restrictions.

On each tetrahedron choose four local heights $h_0, h_1, h_2, h_3 \in \mathbb{Z}$ such that

$$z(i \rightarrow j) = h_j - h_i.$$

The face equations are exactly what permits these local lifts. On a paired face the heights on one

side differ from their transported counterparts by a single integer constant. Thus the local affine functions define a piecewise-linear map to $\mathbb{R}/\mathbb{Z}$. Its inverse image of $\frac{1}{2} + \mathbb{Z}$ is a properly embedded, cooriented normal surface, called the *coherent fibre* of z in $\mathcal{T}$ and denoted $S(\mathcal{T}, z)$. On every oriented edge its intersection signs agree.

Adding a constant to all four heights in a tetrahedron changes no local normal coordinate. Our implementation normalizes the first local height to zero. This convention controls representation size and makes independent replay deterministic; it is not a vertex coboundary.

Lemma 2.1 (Local normal coordinates). *Order the corners so that $h_a \leq h_b \leq h_c \leq h_d$, resolving equalities by the local vertex index. The coherent fibre contains $h_b - h_a$ triangles at a , $h_d - h_c$ triangles at d , and $h_c - h_b$ quadrilaterals separating $\{a, b\}$ from $\{c, d\}$. Every other coordinate is zero. Its number of normal discs in that tetrahedron is*

$$P_t = \text{rng}(h_0, h_1, h_2, h_3) := \max_i h_i - \min_i h_i.$$

Proof. A half-integer level between the lowest and the second-lowest height cuts off the lowest corner. A level between the middle heights separates two corners from two, and a level between the second-highest and highest height cuts off the highest corner. These three disjoint sets of levels have the stated cardinalities. Tied heights create zero-length ranges and require no exception. Their sum is the total height range. $\square$

The construction handles exponentially many normal discs by a constant number of integers per tetrahedron. It is nevertheless only one class of normal surfaces. In particular, it suppresses pairs of oppositely signed edge intersections; such a pair can occur in a topology-preserving normal representative of the same cohomological data.

2.2 Cohomology and a change of gauge

For an integral vertex potential p , the cocycle $z + \delta p$ has local heights $h_i + p(v_i)$. It represents the same cohomology class as z but can give a different coherent normal fibre. The existing project uses a maximal-tree gauge to extract a primitive rank-one class, and can then optimize the total number of discs over vertex potentials. Neither primitivity alone nor an arbitrary change of gauge should be confused with a proof that a subsequently transported fibre is connected.

Our positive consumer does not need a connectivity invariant throughout the search. It checks a terminal normal vector with the existing component kernel and requires that an essential disc occur as a component. This permits the transported fibre to split or to contain additional components.

2.3 The formal Pachner region

Give a triangular bipyramid belt labels C, D, E and apex labels A, B . The two-tetrahedron filling is

$$\mathcal{B}_2 = \{ACDE, BCDE\}.$$

The three-tetrahedron filling is

$$\mathcal{B}_3 = \{ABCD, ABDE, ABEC\}.$$

The six boundary triangles agree. A 2–3 move replaces $\mathcal{B}_2$ by $\mathcal{B}_3$ and creates the interior edge AB ; the inverse 3–2 move removes that edge.

A valid site must satisfy the full formal-region gluing conditions. Edge degree three alone is insufficient. The native 3–2 checker requires three distinct tetrahedra, propagates the formal labels through their interior faces, verifies the three expected vertex sets, and checks every boundary face correspondence. Global boundary identifications of the formal region are allowed by the existing checked contract. The interior cell replacement is what matters for the argument below.

3 Exact jumps across a bipyramid

3.1 Five heights and preservation of the class

Lemma 3.1 (Local cochain transport). *An integral cocycle on either filling of a valid Pachner region determines five heights H_A, H_B, H_C, H_D, H_E , up to an additive constant, by lifting through the formal region. Restricting these heights to the other filling defines a cocycle that agrees on every unchanged edge and represents the same integral cohomology class under the move’s relative-boundary identification. Computing the five heights takes a constant number of integer operations once the labelled region has been supplied.*

Proof. Choose one local tetrahedron lift. On a shared interior face, all differences of the three shared corner heights are prescribed by the cocycle, so the lift on its neighbour is fixed by one additive adjustment. Propagation through the two- or three-tetrahedron formal ball yields values on the five labels. The cocycle condition makes different propagation routes consistent; the implementation checks this consistency rather than assuming it.

The new edge values are differences of the same five values. They therefore satisfy all new face equations and retain the values on the six boundary triangles. On the formal ball both cochains are coboundaries of a local height function. These functions agree at the boundary vertices, and their affine extensions are homotopic relative to that boundary. Gluing this local homotopy to the unchanged exterior shows that the two circle-valued maps represent the same class. Equivalently, evaluate the cocycle on a cycle and replace every segment crossing the ball by a boundary path; both evaluations agree. The finite region has at most three tetrahedra, so only a constant number of additions, subtractions, and comparisons occur. $\square$

This lemma transports the cocycle. It makes no assertion that the fibres before and after the move are isotopic.

3.2 Closed formulas

Sort the belt heights and the apex heights independently:

$$x \leq y \leq z \quad \text{are } H_C, H_D, H_E, \quad L \leq U \quad \text{are } H_A, H_B.$$

Set

$$g = [L - z]_+ + [x - U]_+, \tag{1}$$

$$\Delta P = \max(U, y) - \min(L, y) + [\min(U, x) - L]_+ + [U - \max(L, z)]_+. \tag{2}$$

Here $[r]_+ = \max(0, r)$. The number g is the gap between the closed interval of belt heights and the closed interval of apex heights, taken as zero when the intervals meet.

Theorem 3.2 (Exact coherent-fibre jumps). *For cochain transport through a valid 2–3 move,*

$$\chi(S(\mathcal{T}_3, z_3)) - \chi(S(\mathcal{T}_2, z_2)) = -2g, \quad (3)$$

$$P(\mathcal{T}_3, z_3) - P(\mathcal{T}_2, z_2) = \Delta P \geq U - L \geq 0. \quad (4)$$

The inverse 3–2 move has Euler gain $2g$ and normal-disc saving ΔP . The statements concern the total coherent fibre, including all its components. They remain valid with arbitrary integral heights, tied heights, independent local additive offsets, and boundary identifications allowed by the checked formal-region contract.

Proof by relative cell counts. Write $r(a_1, \dots, a_k) = \max_i a_i - \min_i a_i$. Lemma 2.1 gives

$$\begin{aligned} P_2 &= r(L, x, y, z) + r(U, x, y, z), \\ P_3 &= r(L, U, x, y) + r(L, U, y, z) + r(L, U, z, x). \end{aligned}$$

Using $x \leq y \leq z$ and $L \leq U$ and collecting the extrema yields

$$\begin{aligned} P_3 - P_2 &= \max(U, y) - \min(L, y) \\ &\quad + \min(U, x) - \min(L, x) + \max(U, z) - \max(L, z). \end{aligned} \quad (5)$$

The difference of minima is $[\min(U, x) - L]_+$, and the difference of maxima is $[U - \max(L, z)]_+$. This proves (2). The first term is the range of $\{L, U, y\}$ and is at least $U - L$; the other two terms are nonnegative.

For Euler characteristic, the unchanged boundary cell contributions cancel. The old internal face CDE contributes $r(x, y, z) = z - x$ normal arcs. The new three internal faces contribute

$$r(L, U, x) + r(L, U, y) + r(L, U, z).$$

The new internal edge AB contributes $U - L$ intersection points. There are no new interior vertices. Thus

$$\begin{aligned} \Delta\chi &= \Delta P - r(L, U, x) - r(L, U, y) - r(L, U, z) \\ &\quad + (z - x) + (U - L). \end{aligned}$$

Substitution of (5) cancels the terms containing the median y and gives

$$\Delta\chi = -|U - x| - |L - z| + (z - x) + (U - L).$$

Use $|U - x| = (U - x) + 2[x - U]_+$ and $|L - z| = (z - L) + 2[L - z]_+$ to obtain (3).

This proof is relative to the formal boundary, so identifications among its cells do not change the cancellation argument. Only the replaced interior faces and edge contribute to the difference. The reverse direction negates both jumps. $\square$

3.3 An independent coarea check

At a half-integer level, let $a \in \{0, 1, 2\}$ be the number of apices above the level and $b \in \{0, 1, 2, 3\}$ the number of belt vertices above it. The following table gives $(\Delta\chi, \Delta P)$ for the two-to-three replacement at that single level.

	$b = 0$	$b = 1$	$b = 2$	$b = 3$
$a = 0$	(0, 0)	(0, 0)	(0, 1)	(-2, 1)
$a = 1$	(0, 2)	(0, 1)	(0, 1)	(0, 2)
$a = 2$	(-2, 1)	(0, 1)	(0, 0)	(0, 0)

For example, when both apices are above all belt vertices, each old tetrahedron contributes a triangular disc, while the three new quadrilaterals form an annulus around the new edge. There are exactly $L - z$ such levels. The reversed height ordering contributes $x - U$ levels. Summing gives the gap formula. In other nonempty single-level cases the local surface is a disc, although its subdivision and the number of pieces can change.

The table provides a second way to test the closed formulas and handles ties without numerical limits. It is not an excuse to enumerate levels: the implementation uses the five integers directly. In particular, 20,000-bit heights do not trigger 2^{20000} iterations.

Several exceptional annuli may be nested. We do not infer a family of simultaneously disjoint compression discs for the entire original surface from the single-level picture. The relative cellular proof already gives the claimed global formula without this stronger geometric assertion.

3.4 The existing obstruction as a numerical example

In the repository's eight-tetrahedron example, one eligible collapse has formal heights

$$(H_C, H_D, H_E, H_A, H_B) = (0, -1, 1, 3, 4).$$

Consequently $(x, y, z) = (-1, 0, 1)$ and $(L, U) = (3, 4)$. The gap is $g = 2$, the Euler gain of the 3-2 move is 4, and

$$\Delta P = (4 - 0) + 0 + (4 - 3) = 5.$$

A checked downward epoch carries the original cochain through six collapses, ending with two tetrahedra and an eight-piece normal disc. The disc is established by the component certificate, not by substituting the Euler gain into a connected-surface genus formula. The retained record includes the preceding solid-torus construction; this particular fixture is not presented as a witness for an independently supplied knot diagram. Separate diagram-bound examples are tested below.

3.5 Connectivity can fail: an explicit checked example

The archive includes a five-to-four-tetrahedron collapse whose incoming primitive coherent fibre is one compressing disc with 24 normal pieces. Its five heights, in belt-then-apex order, are

$$(0, 2, 1, 5, 4).$$

Thus $(x, y, z) = (0, 1, 2)$ and $(L, U) = (4, 5)$, giving $g = 2$, Euler gain 4, and piece saving 5. The outgoing fibre is one 11-piece compressing disc together with two four-piece spheres. Its Euler characteristic is $1 + 2 + 2 = 5$ and its total piece count is 19. Both component censuses have independently replayed certificates.

The input arises from the existing interior-vertex solid-torus fixture, an integral vertex coboundary, and one checked expansion. The coboundary and transport preserve the primitive cohomology class. This provides a direct counterexample to preservation of fibre connectivity under a legal collapse, and to the proposed use of $\chi \leq 1$ for every transported primitive fibre. It also explains why the source consumer uses the general component kernel even when the initial seed was connected.

Example 3.3 (Large geometric changes with short binary descriptions). With $x = y = z = 0$ and $L = U = H > 0$, the upward move has $\Delta\chi = -2H$ and $\Delta P = H$, although the new edge has cocycle value zero. With $L = -H$, $U = H$, and the same belt, $g = 0$ but $\Delta P = 4H$. These changes require only $O(\log H)$ input bits. Neither unit Euler progress nor literal piece count is an appropriate polynomial iteration budget; the formulas operate on encoded integers.

4 Scored descent and binary complexity

4.1 One scan for all collapse scores

The candidate scanner prepares the manifold once, groups its $6t$ local edge occurrences by signed global edge class, and examines every degree-three class. A class is accepted only when its three tetrahedra and interior face maps form a valid labelled bipyramid. Each such test uses constant-size local data. The five heights and the two scores are then obtained from Theorem 3.2.

The score policy chooses the largest Euler gain, then the largest disc saving, with remaining ties resolved by local incidence order. A reference policy chooses the first valid site. Every accepted move is constructed by the existing Pachner producer and then checked independently. Every cocycle transport is also checked independently using signed edge values and global normal cell counts; the transport verifier does not call the score formula.

Proposition 4.1 (Candidate-discovery cost). *After finite-manifold preparation, all eligible 3–2 candidates and their scores can be produced using $O(t)$ indexed operations and $O(t)$ integer operations on the input edge values. A score-based choice requires one further linear scan. The candidate operation neither solves cohomology nor expands a normal surface.*

Proof. There are $6t$ local edge occurrences. Grouping visits each once. A rejected group with a degree other than three is discarded by its recorded size; each remaining group contains exactly three tetrahedra and at most twelve face records. Label propagation, consistency checking, height lifting, and score evaluation therefore take constant work per candidate. There are at most $2t$ degree-three groups, which proves the bound. $\square$

The statement counts indexed operations after the existing preparation routine. Deterministic balanced dictionaries can replace expected-time hash tables at an additional logarithmic factor. Python container costs, certificate materialization, and the independent geometric checks appear in the measured implementation time and are not hidden inside the constant-size score claim.

4.2 A complete downward epoch

An epoch repeatedly discovers the eligible sites, chooses a site by the fixed policy, performs the collapse, and transports the cochain. A move limit can stop it early; the resulting transcript remains valid, but an early stop is not declared a local minimum.

Theorem 4.2 (Downward-epoch bounds). *An uncapped 3–2 epoch starting with t_0 tetrahedra contains fewer than t_0 successful moves. Let*

$$D_0 = \max_e |z_0(e)|, \quad B_0 = \lceil \log_2(1 + D_0) \rceil.$$

At every stage the edge max-norm is at most D_0 , each normalized local height has absolute value at most D_0 , and every normal coordinate is at most D_0 . The total normal-disc count never increases and the total Euler characteristic never decreases. The entire epoch has polynomial encoded cost in $t_0 + B_0$, assuming the finite-manifold preparation and replay operations used at a supplied state have their stated polynomial cost.

Proof. Every successful collapse removes one tetrahedron. A 3–2 move removes the edge AB and creates no edge; every new tetrahedral edge is an old boundary edge of the bipyramid. Lemma 3.1 preserves its cocycle value. Hence the maximum absolute edge value does not increase. After setting the first corner height to zero, the other heights are edge differences from that corner. Each coordinate in Lemma 2.1 is a difference along another tetrahedral edge, so it has the same bound. The two monotonicity statements follow from Theorem 3.2.

There are $O(t_0)$ stages, each with $O(t_0)$ candidate work and a constant number of supplied-state preparation, rewriting, and replay operations. All arithmetic integers have $O(B_0 + \log t_0)$ bits, including the sums in global cell counts. Thus their composition is polynomial. If all preparations are implemented by the native union–find and indexed cell passes, the indexed work is $O(t_0^2 \alpha(t_0))$ apart from dictionary overheads and integer-bit costs. Retaining every full triangulation and height array gives a quadratic-size transcript; the present code does retain these arrays and claims no persistent-data-structure improvement. $\square$

The theorem proves efficient completion of a geometric epoch. It does not assert that the final coherent fibre contains a compressing disc. In particular, Euler monotonicity is a statement about a sum over components, not a connectedness theorem.

These complexity bounds use normalized initial height rows, or an edge-value encoding of the cocycle. Arbitrary supplied additive offsets can have many more bits than the edge differences. Their actual input length must be included in the one-time reading and normalization cost. The identities remain offset-invariant, and all subsequent transported rows satisfy the advertised normalized-height bound. A quadratic-size transcript has $O(t_0^2)$ integer slots, or $O(t_0^2(B_0 + \log(t_0 + 1)))$ bits including incidence indices.

4.3 Upward moves and a precisely stated conditional search bound

Theorem 4.3 (Bit growth charged to upward moves). *Consider any sequence consisting of transported 2–3 and 3–2 moves, starting with (t_0, z_0) , and containing u upward 2–3 moves in total. At every prefix,*

$$t \leq t_0 + u, \quad D \leq 2^u D_0, \quad B \leq B_0 + u.$$

Accordingly, its current cocycle and normal-coordinate arrays use $O((t_0 + u)(B_0 + u + 1))$ bits, with an additional $O((t_0 + u) \log(t_0 + u + 1))$ bits for face-pairing indices.

Proof. Only an upward move can increase the number of tetrahedra. Its new edge value satisfies

$$z'(A \rightarrow B) = z(A \rightarrow C) + z(C \rightarrow B)$$

for any belt vertex C , so its absolute value is at most $2D$. Every other edge value is retained. Downward moves do not increase D . Induction gives the first two bounds. Since $1 + 2^u D_0 \leq 2^u(1 + D_0)$, the ceiling-logarithm bound follows. There are four height and seven normal-coordinate entries per tetrahedron, and Lemma 2.1 bounds each by the current D after normalization. $\square$

This bound applies to the transported cochain. An algorithm that re-solves cohomology and chooses a new vertex gauge introduces an additional operation whose magnitude must be analysed separately. It also does not bound the number of intermediate states visited by a search.

A path containing u upward and d downward moves has final size $t_0 + u - d \geq 1$. Therefore $d \leq t_0 + u - 1$, and its total number of moves is at most $t_0 + 2u - 1$. Storing a full state at every step uses $O((t_0 + u)^2)$ indexed records, each with $O(B_0 + u + \log(t_0 + u + 1))$ bits. This is a path-size bound. It does not bound the number of possible paths, because many downward choices may branch at each stage.

Fix a deterministic maximal downward policy $\mathcal{D}$. A *macrostate* is its output. A macrostep chooses one valid upward move and then applies $\mathcal{D}$ again. Since a triangulation with t tetrahedra has at most $2t$ paired faces, it has at most $2t$ candidate upward sites of this kind.

Corollary 4.4 (A bounded-escape macrofamily). *Exhaustively searching this specified macrofamily to upward depth u visits at most*

$$\sum_{j=0}^u (2(t_0 + u))^j$$

macrostate tree nodes. Intermediate triangulations inside each deterministic downward epoch are charged to that node's polynomial cost. With polynomial complete supplied-state disc-component queries, the total cost is

$$(2(t_0 + u) + 1)^{u+1} \text{poly}(t_0 + u, B_0 + u).$$

For $t_0 = \text{poly}(n)$, $B_0 = \text{poly}(n)$, and $u = O(\log n)$ this is $n^{O(\log n)}$. The completeness hypothesis needed for an input class is that every unknotted input in that class has at least one macrostate in this particular deterministic macrofamily, within that depth, whose coherent fibre contains an essential-disc component.

Proof. Theorem 4.3 bounds every encountered triangulation and integer array. The number of children is at most $2(t_0 + u)$ and each downward epoch has polynomial cost by Theorem 4.2. Sum the breadth of the depth- u tree and apply the query bound. $\square$

This is a mathematical reduction of the search target, not a completed recognition theorem or an implemented exhaustive macrosearch. Different downward choices can matter. Replacing all downward choices by one greedy normal form without a coverage argument would be an invalid use of the corollary. Nor does the existence of a short arbitrary Pachner path imply the needed normal-form path bound. The shipped search implements a single downward epoch; the corollary specifies a testable extension and its exact remaining assumption.

Remark 4.5 (Allowing bursts of upward moves). The single-upward-move family is particularly restrictive: immediately after an expansion its inverse collapse is valid, so the maximal downward epoch performs at least one collapse. Macrostate tetrahedron counts can never increase, and the global size peak is at most $t_0 + 1$ after the initial closure. This family cannot represent an escape that requires two consecutive expansions before any downward simplification.

A more flexible, still precisely bounded family lets each upward move carry one additional decision: continue the current upward burst, or close it by applying $\mathcal{D}$. Every nonempty accepted burst ends with a close decision; the initial $\mathcal{D}$ -state is also queried. At most u upward choices and u binary closure choices occur on a branch. There are at most $4(t_0 + u)$ children per tree node, so

$$\sum_{j=0}^u (4(t_0 + u))^j \text{poly}(t_0 + u, B_0 + u)$$

bounds the work. This is again $n^{O(\log n)}$ for $u = O(\log n)$ and polynomial initial size. No zero-length bursts are allowed. A witness coverage theorem must now concern this larger, explicitly specified family; it is still not a consequence of an arbitrary short Pachner path. The burst variant is a research formulation, not shipped search code.

5 Ordered boundary data from compressed curves

5.1 The information that component inventories lose

Four marked occurrences a, b, c, d can lie on the same circle in the cyclic order (a, b, c, d) or (a, c, b, d) . Their component-incidence vectors agree, but gluing bands to those occurrences can produce different boundary patterns. A hierarchy interface must therefore retain occurrence order, not just the fact that the four marks are connected.

The following problem is intentionally narrower than constructing an entire cut manifold. Its input is a degree-two graph already encoded by interval pairings and a finite set of explicit marked vertices. Its output is the first-return pairing between marked half-edges and the resulting cyclic words. Actual normal boundary curves supply such inputs through the existing native normal-arc compiler. A future three-dimensional parallelity bundle compiler must supply analogous ordered interface curves and their side labels.

5.2 Multigraph and occurrence conventions

Definition 5.1 (Compressed boundary multigraph). Let $V = [0, N) \cap \mathbb{Z}$. An affine interval pairing is a bijection

$$f : [a, b] \cap \mathbb{Z} \longrightarrow [c, d] \cap \mathbb{Z},$$

with slope $+1$ or -1 and $b - a = d - c$. For every domain point q , the row contributes an undirected edge with endpoints $q, f(q)$. Distinct rows or domain points define distinct edges, even when their endpoints coincide. A loop counts twice toward degree. A compressed boundary graph is a list of such rows whose multigraph has degree two at every vertex.

Every connected finite degree-two multigraph is a circle, including a one-vertex loop and a two-vertex circle with parallel edges. It is essential to validate degree in this multigraph before passing to an orbit relation: the orbit relation alone discards edge multiplicities and cannot validate the hypothesis.

For k pairing rows, degree validation is an endpoint-event sweep. Each domain and range contributes $+1$ at its inclusive lower endpoint and -1 one past its upper endpoint. The running count must equal two on every nonempty segment of $[0, N)$. This uses $O(k \log(k + 1))$ comparisons of binary endpoints and never visits all N vertices.

Let p distinct marked vertices, forming a set M , be supplied in an explicit list. Their positions in that list are occurrence identifiers; repeated display names may be maintained by the caller, but duplicate geometric points are rejected. A marked half-edge has the exact identity

$$(\text{pairing-row index, original domain point, endpoint side } 0 \text{ or } 1).$$

This distinguishes both ends of a self-loop and parallel edges. There are exactly $2p$ marked half-edges.

5.3 Deleting marks without expanding the curve

For each row, delete domain points that are marked, and preimages of marked range points. Split the remaining domain into maximal integer intervals. Then close the p deleted gaps in the universe by the rank map

$$\rho(q) = q - \#\{m \in M : m < q\}, \quad q \notin M.$$

On every surviving domain and range interval the rank map is a translation; the induced pairing is therefore again affine with the original sign.

An original edge whose two endpoints are marked is recorded directly as a zero-interior-length connection. Every other marked half-edge has one unmarked neighbour; record that neighbour's compressed rank and the half-edge label. The remaining graph has $N - p$ vertices. It consists of paths with two missing endpoint incidences, and circles with no missing incidence. An isolated unmarked vertex between two marks is a path of length one in this convention and is included.

Lemma 5.2 (Residual-size bound). *The residual graph has at most $k + 2p$ interval-pairing rows. The number of endpoint annotations is at most $2p$, independently of N .*

Proof. The total number of edge occurrences incident to marks is at most $2p$, because the marked degree sum is $2p$ and a marked-to-marked edge is counted twice there. Removing r_i distinct domain points from row i leaves at most $r_i + 1$ intervals. Hence the total is at most $k + \sum_i r_i \leq k + 2p$. Each nondirect marked incidence produces exactly one endpoint annotation. $\square$

This sharper bound follows from degree two. The implementation uses binary searches in the sorted mark list to find the relevant domain and range marks; it does not need to scan every mark against every integer point.

5.4 Four exact coordinates instead of one coordinate per port

Assign the marked half-edges the distinct positive integer labels $1, \dots, 2p$. For an unmarked vertex v , let $E(v)$ be the labels of the missing incidences at v . Give v the additive weight

$$w(v) = \left(|E(v)|, \sum_{a \in E(v)} a, \sum_{a \in E(v)} a^2, 1 \right). \quad (6)$$

The last coordinate is one on the entire residual universe. The other coordinates are supported at at most $2p$ annotated vertices. Several annotations at one vertex add exactly.

Theorem 5.3 (Exact two-endpoint decoding). *Let (c, s, q, ℓ) be the orbit sum of (6) for a component of the validated residual graph.*

1. *If $c = 0$, then $s = q = 0$ and the component is an unmarked circle with ℓ vertices.*
2. *If $c = 2$, the component is a path with two distinct marked half-edge labels a, b . They are uniquely determined by*

$$d^2 = 2q - s^2, \quad a = \frac{s - d}{2}, \quad b = \frac{s + d}{2}, \quad a < b.$$

The path has ℓ unmarked vertices. Its multiplicity in the orbit histogram is one.

There are no other cases. Thus four additive coordinates recover the same endpoint matching and gap lengths as the $2p + 1$ -coordinate vector consisting of an indicator for every port and one length coordinate.

Proof. Every residual component came from a circle after deleting some marked vertices. If it meets no deleted incidence, it is a whole original circle. Otherwise it is a path between two consecutive marked incidences, so it has exactly two missing endpoint occurrences. These occurrences are distinct half-edges, even when their marks are the same or the residual path is a single vertex. Since labels are globally unique, their orbit sum satisfies $c = 2$, $s = a + b$, and $q = a^2 + b^2$. Therefore

$$2q - s^2 = 2(a^2 + b^2) - (a + b)^2 = (a - b)^2.$$

The nonnegative square root and the sum determine the unordered pair. Two distinct path components cannot have the same pair of labels, since each annotation occurs once, so a marked-path histogram record has multiplicity one. The fourth coordinate counts its residual vertices. Unmarked circles with the same length can and should be aggregated by multiplicity. $\square$

The decoder checks that c is zero or two; in the second case it checks that the discriminant is a positive exact square, that $s \pm d$ are even, and that $1 \leq a < b \leq 2p$. It also checks that every port is used exactly once after the direct connections are included. No floating-point square root is used.

This argument depends on a proved two-endpoint bound. The first two moments do not in general identify arbitrary multisets. For example, $\{1, 5, 6\}$ and $\{2, 3, 7\}$ have the same size, sum, and sum of squares. Consequently the four-coordinate decoder is not a general replacement for all component-incidence vectors. Its exactness is supplied by the validated boundary topology.

Corollary 5.4 (Size of the moment query). *The query has dimension four, at most $k + 2p$ pairing rows, and at most $2p + 1$ initially supplied weight intervals. Its nonlength coordinates have $O(\log(p + 1))$ bits under nonnegative weight transport, and its length coordinate has $O(\log(N + 1))$ bits. In particular, the moment encoding does not hide a linear number of endpoint labels inside one exponentially long integer.*

Proof. Initially there are at most $2p$ endpoint annotations and one constant length interval. The total first coordinate is at most $2p$, the total label sum is at most $\sum_{a=1}^{2p} a = O(p^2)$, and the total square sum is at most $\sum_{a=1}^{2p} a^2 = O(p^3)$. All are nonnegative. Every stored transported contribution is bounded by the respective conserved total. The total length is $N - p$. These totals have the stated binary lengths. The pairing bound is Lemma 5.2. $\square$

Combined with the weighted AHT theorem, this gives polynomial theoretical time in the compressed input size and p . If overlapping supplied weight intervals are converted into disjoint constant runs, the at most $2p$ annotated points give at most $4p + 1$ runs, still $O(p)$; this is the piecewise-constant weight-size parameter relevant to the orbit theorem. The independently replayed implementation is separately tested; the experiments do not constitute a new formal proof of the orbit engine's running-time theorem.

5.5 Recovering cyclic words and exact gaps

The decoded connections define a fixed-point-free involution α on the $2p$ marked half-edges, pairing the ends of every gap. At each mark, the two incident half-edges define another fixed-point-free

involution τ . Starting from an outgoing half-edge h , travel across a gap by α and depart the next mark by τ . Thus $\tau\alpha$ traces one orientation of an original marked circle.

For each circle the default convention starts at its smallest occurrence identifier and chooses the lexicographically smaller of the two directions. A tie in a one- or two-mark circle is resolved by the outgoing half-edge word, so the two sides of a loop remain distinguishable. The caller can instead supply an explicit marked half-edge to fix the start and direction on its own component.

Each output gap records the number ℓ of *unmarked vertices* between successive marks. It therefore contains $\ell + 1$ original edges. An entire marked cycle with r occurrences contains $r + \sum_i \ell_i$ vertices. The weighted replay preserves the residual mass $N - p$, so reconstruction implies that marked cycle lengths plus unmarked cycle lengths, counted with multiplicity, sum to N . Tests also check this global equality explicitly.

The output includes exact half-edge descriptors, the gap matching, canonical marked cycles, and a length–multiplicity histogram for circles with no marks. It deliberately retains the latter: discarding a large unmarked family would destroy the topology of a later cut representation.

5.6 What remains outside this interface

The marked graph does not by itself identify which side is horizontal in an I -bundle, transport a meridian–longitude basis, or construct all nonparallel pieces of a three-dimensional cut. A two-sheeted signed cover is a natural way to extend the endpoint query to side parity, but that extension is not claimed as shipped functionality here. Likewise, two attachments at the same geometric point need a specified local ordering; the current API rejects duplicate marked points.

Lackenby’s compressed cutting theorem supplies the theoretical context for these additional data. The present normal-boundary adapter exercises the same ordered-curve primitive on actual normal arcs. It is a concrete interface milestone, not the complete implementation of that theorem.

6 Certificates and the source-bound positive search

6.1 Separate mathematical responsibilities

The implementation makes the following sequence checkable:

1. Reconstruct a valid classical one-component planar diagram and verify its canonical compact exterior against the existing independent geometric checker.
2. If preliminary boundary shellings were requested, verify the exact removed tetrahedra and the resulting face pairings.
3. Validate the initial integral height rows and their signed global edge consistency.
4. For every transcript step, verify the Pachner replacement, the unchanged exterior, the five-height lift, the new height rows, the coherent normal coordinates, and both global cell-count jumps.
5. Independently reconstruct the final coherent normal vector, replay its compressed disc-component certificate, and require a strictly positive number of essential disc components.

Here an essential disc is a properly embedded disc whose boundary is essential in the boundary

torus. The final implication is the standard one: the boundary of a knot exterior is compressible exactly for the unknot [5, discussion of Corollary 1.2]. Its use is justified here only after the correspondence with the supplied diagram and every intervening homeomorphism have been checked. A valid zero-disc certificate is rejected as an unknot witness.

6.2 The experimental positive stage

`transport_seed_decide` constructs the canonical exterior, optionally performs the existing boundary shellings, extracts one primitive integral cocycle, optionally selects its existing minimum-span representative, and performs one deterministic downward epoch. It submits the final vector to the general compressed disc-component kernel. The return value is `UNKNOT` with a complete certificate, or `INCONCLUSIVE`. It never returns `KNOTTED` from failed coherent-fibre search.

The stage is an additive experimental API. It is not enabled in the maintained recognition schedule. The move cap, shared work allowance, and orbit-cycle allowance are explicit. The work allowance counts cooperative guard invocations, not machine instructions or bit operations. All stages, including positive certificate replay, share that allowance. Exceptions raised by the caller's cancellation callback propagate.

6.3 Independent replay and its limits

The cocycle transport checker imports neither a transport producer nor a cocycle solver. It does not call the score formula: it reconstructs signed edge values and computes the old and new normal Euler characteristics from the cell structure. The source consumer imports independent exterior, shelling, move, and component checkers. Tests patch the discovery and producer entry points to raise exceptions while the consumer continues to verify a positive proof.

The marked-order checker independently reconstructs the degree-two source graph, the deleted-mark instance, the endpoint weights, and the cyclic solution, and then replays the supplied weighted orbit trace. Its certificate is bound to the ordered pairing rows and the original mark list; relabelling or reordering those inputs without changing the certificate is not accepted.

Shared low-level integer and finite-geometry routines remain part of the trusted code base. These checkers are independent algorithmic replays, not a Lean formalization or a separately verified implementation of every supporting library. The archive records the exact code and test scope so that this distinction is reviewable.

7 Reproducible experiments

7.1 Environment, baselines, and units

The experiments use CPython 3.12.14 on Linux x86-64. Independent native comparisons use the Regina Python distribution 7.4.1, whose engine reports version 7.4 [6]. The archive contains the source inputs, certificate-bearing records, benchmark drivers, and an environment record. All times in the tables are wall-clock milliseconds on the research machine; they are measurements of the stated workloads, not complexity exponents fitted from a few points.

The all-candidate scoring comparison uses five trials per arm, alternating which arm runs first. The final endpoint-encoding comparison uses three alternating trials per arm on identical normal

curves and mark lists. Raw samples are retained for those paired comparisons. The literal curve walk is capped at 200,000 represented vertices; omitted larger walks are not timed or extrapolated. The initial exploratory records are retained separately and are not substituted for the matched comparison.

The machine is shared. Controlled measurements were run with other heavy research jobs paused, but hardware isolation and a distribution over machines are not claimed. Median ratios are descriptive. They do not include uncertainty estimates, and the complete recognizer may spend most of its time in a different operation.

7.2 Exact score checks and independent native moves

The local audit evaluates all $5^5 = 3,125$ five-height assignments in $\{-2, -1, 0, 1, 2\}^5$. It compares the formulas with separate tetrahedron, face, and edge range counts. The regression also scales heights by 2^{20000} and confirms unchanged deterministic guard counts, while the arithmetic integers themselves become longer. This checks the absence of a loop over represented levels; it is not a claim that big-integer arithmetic has constant cost.

A seeded mixed-move audit performs 120 moves: 63 upward and 57 downward. The four starting inputs are layered solid tori with two and eight tetrahedra, a one-crossing diagram exterior, and a trefoil exterior. Each accepted move passes independent cocycle replay. Regina's separate Pachner operation gives an isomorphic replacement triangulation, and its normal-surface Euler characteristic agrees with the native cell count. The observed edge-magnitude ratio attains two in an upward move, so the factor-two bound cannot be uniformly improved for this max-norm.

The eight-tetrahedron obstruction and the connectivity counterexample from Section 3.5 are retained with complete construction data and component certificates. The former finishes after six collapses with two tetrahedra and one eight-piece compressing disc, saving 33 pieces overall. The latter verifies the split into a disc and two spheres. Neither fixture is mislabeled as a proof about an unrelated diagram.

7.3 All eligible collapse scores: a matched comparison

The benchmark begins with layered solid tori of sizes 16, 32, 64, 128, 256. Expanding disjoint pairs gives the sizes and eligible collapse counts in Table 1. These triangulations have one global vertex. Therefore two primitive cocycles for the rank-one group differ only by overall sign: there are no nonzero vertex coboundaries. The coherent normal vector is unchanged by that sign. This makes fresh cohomology extraction and direct transport comparable on exactly the same geometric outputs.

The reference arm first recognizes eligible sites without scoring them. For every site it constructs the full replacement triangulation, extracts a fresh primitive rank-one cocycle, and computes its Euler characteristic and normal-disc count. The new arm obtains every score directly from the input cochain in one scan. The driver requires equality of all per-site Euler and piece outputs before collecting timings.

At the largest measured input, direct scoring takes 20.519 ms versus 6857.931 ms for the fresh-extraction arm, a 334.23-fold ratio. This is not a 334.23-fold speedup of unknot recognition.

The comparison also has an asymptotic explanation independent of the timings. There are $\Theta(t)$ candidate sites in this family. The reference arm explicitly materializes a whole $\Theta(t)$ -size triangulation for each site, so those writes alone cost $\Omega(t^2)$ indexed operations. The new scoring

t	Sites	Bits	Local (ms)	Fresh H^1 (ms)	Ratio
24	8	13	0.933	18.597	19.93×
48	16	24	1.894	71.151	37.57×
96	32	46	3.855	300.348	77.92×
192	64	90	8.259	1 300.017	157.40×
384	128	179	20.519	6 857.931	334.23×

Table 1: Scoring all eligible collapses. Times include each arm’s stated preparation and discovery work. The speedup is the ratio of median times. Height bits grow with the layered family.

stage uses $O(t)$ integer operations after preparation. Cohomology elimination, integer-bit costs, and final successful-certificate construction are additional costs with their own bounds.

7.4 Diagram corpus: unchanged coverage is a result

The maintained source audit supplies 84 source cases. The experiment selects the 82 with at most sixteen crossings, reconstructs each diagram’s canonical compact exterior, and compares three supplied-coherent-vector strategies:

1. the first eligible collapse followed by a fresh tree-gauge cocycle;
2. the first eligible collapse with the original cochain transported;
3. the maximum Euler-gain, then maximum piece-saving collapse with the original cochain transported.

The shared descent-and-query allowance is two million cooperative guard calls per arm, with a terminal orbit limit of 5,000 cycles. The initial canonical exterior and seed construction precede this measured arm budget. The separate source API charges its own full pipeline to a shared budget.

All 246 arm runs complete. Every arm performs 144 collapses in total and finds 15 positive disc-component cases. All 30 positive transport runs pass the independent diagram/exterior/transport/component consumer. Thus this experiment shows no additional recognition coverage.

Final Euler characteristics agree across all three arms on all 82 sources. The two transport policies also have identical final piece counts on every source. Fresh gauge extraction changes that count on twelve sources: its total is 16,170, compared with 16,297 in either transport arm. Accordingly, these data do not establish that the score ordering produces better final geometry than first-site transport on the ordinary diagram corpus. They demonstrate correct reuse and identify a workload where the proposed priority is neutral.

The corpus record contains timings for diagnosis, but they are not a matched speed comparison. In particular, full source replay was included for positive transport runs, and the exploratory collection briefly overlapped another audit. The scoring table above is the controlled performance evidence. Positive proofs are replayed again from the packaged records; source cases whose coherent query fails remain inconclusive rather than being classified as knotted.

7.5 Marked-order correctness and endpoint-encoding performance

The marked-order feature tests compare 500 seeded signed interval-exchange multigraphs with a literal edge-occurrence oracle. One hundred further cases compare both weight encodings on the

same input. Tests include one-vertex loops, reflection fixed points, parallel rows, adjacent marks, singleton residual paths, empty and all-marked graphs, and directed traversal starting at each port of an adversarial example.

The native tests use normal meridians in layered solid tori, boundary caps, and compatible disc/sphere/Möbius mixtures. Eighteen boundary counts are independently compared with Regina. Source pairing order, occurrence order, encoded integer fields, weights, claimed cycle orders, and trace events are mutated in rejection tests. Work limits and delayed callback exceptions are also exercised.

For the paired encoding comparison, the input is a 64-tetrahedron native normal boundary with 89,891,140,425,706 represented vertices and four raw interval pairings. Only the number of explicit marks changes. The moment and one-hot arms receive the same ordered rows and marks and must return the same solution.

Marks	Producer (ms)		Replay (ms)		Replay ratio
	Moments	One-hot	Moments	One-hot	
8	6.482	7.693	9.987	15.086	1.51×
32	32.110	51.145	72.514	260.093	3.59×
128	462.465	710.939	1 062.141	15 035.489	14.16×

Table 2: Matched endpoint encodings on one native boundary. Producer times include certificate recording. Replay is measured separately with the independent consumer. All times are milliseconds.

At 128 marks the weight dimension falls from 257 to four, while both encodings use 4 215 AHT events. Independent replay improves by a factor of 14.16 in this experiment. The serialized certificate sizes are 344 445 bytes and 507 862 bytes, respectively. The retained orbit trace limits the overall size reduction even after dense endpoint vectors are removed.

There is still substantial growth with the number of marks. The output itself has $2p$ ports, deletion creates more interval pieces, and the orbit trace can become longer. The four-coordinate theorem removes a growing dimension; it does not remove all dependence on p or establish a uniform constant-factor advantage.

7.6 Literal controls and a 20,019-bit boundary size

The retained native-boundary experiment with 22 tetrahedra, eight marks, and 150,050 represented vertices takes 2.197 ms for the certified producer and 3.121 ms for independent replay. The geometry-and-order wrapper takes 3.007 ms; this wrapper timing does not include the additional independent replay. A literal native-arc walk takes 149.076 ms. Its ratio is 67.85 against the producer alone, or 28.03 against producer plus replay. These particular medians were collected before the final callback-only verifier correction; the interval algorithm and represented inputs are unchanged. The final paired encoding table was regenerated after that correction.

Small inputs favor literal walking. At 26 vertices the literal walk takes 0.046 ms versus 0.825 ms for the producer; at 1,220 vertices the times are 1.058 ms and 1.495 ms. In 101 interleaved zero-mark controls on the 26-vertex case, moment replay takes 0.07024 ms versus 0.06827 ms for one-hot replay, about 2.9% slower. With one mark the corresponding times are 0.09115 ms and 0.09023 ms. For zero or one mark the reference dimension is one or three, so the four-coordinate representation should not be expected to win universally.

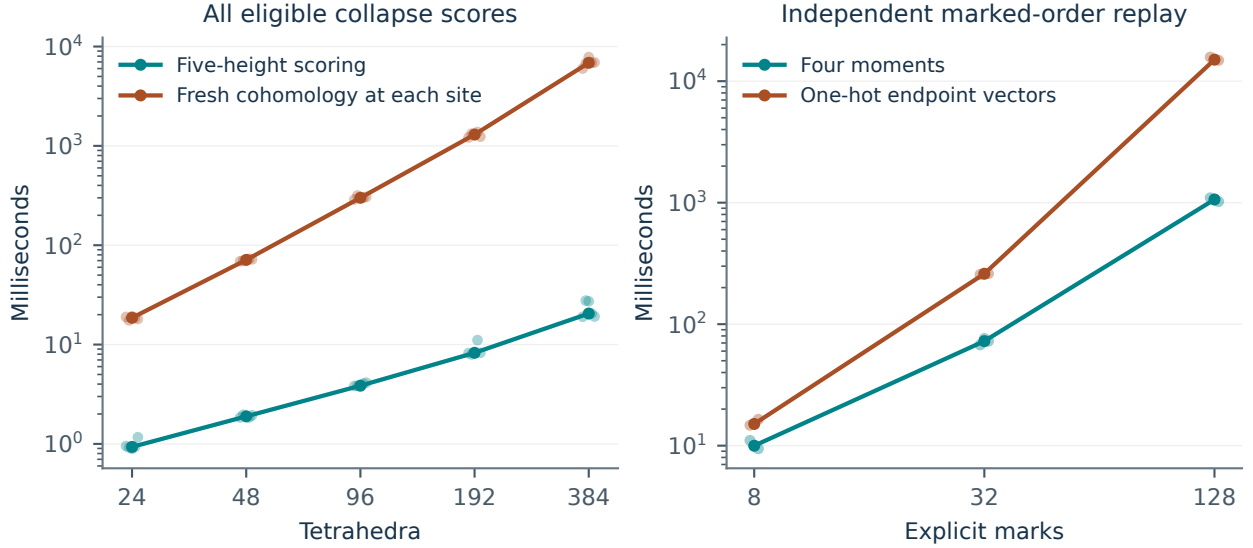


Figure 1: Paired measurements on logarithmic axes. Left: all-candidate scoring; right: independent marked-order replay. Dots show individual retained samples and lines join medians. No asymptotic slope is fitted. Both comparisons require identical mathematical outputs.

A different input multiplies the 24-tetrahedron normal vector by 2^{20000} . The boundary size is

$$N = 392836 \cdot 2^{20000},$$

requiring 20,019 bits. Eight marks leave eight residual pairings from four source rows. The recorded producer time is 8.614 ms, independent replay 12.421 ms, and native-wrapper time 10.158 ms. The certificate has 755,802 bytes and 108 AHT events. Three boundary components contain marks; the remaining untouched family is retained with exact multiplicities, giving 2^{20000} components overall.

This last experiment tests dependence on binary encoding and preservation of multiplicity. It neither expands nor literally checks every represented component, and it is not a benchmark of recognizing a knot with 2^{20000} crossings.

7.7 Regression and retained-proof validation

The packaged snapshot passes 1289 tests in the full `unittest` discovery run, including the existing repository suite and the new features. The retained-proof validation checks the source-bound transport positives, the obstruction transcript, the connectivity counterexample, and the correspondence between computed table entries and their stored data. The machine-readable validation record distinguishes full-suite tests, feature tests, proof replays, and benchmark comparisons.

An initial full-suite attempt exposed a missing historical test oracle in the standalone snapshot, not a mathematical failure. The exact pinned `reports/24/reference/graded_transfer_v1.py` and the other required historical oracle files are now included. The final full-suite run is performed from the deliverable snapshot, with no references to an unshipped checkout.

The validation supports this source revision and these contracts. It is not a proof that all possible triangulations or all possible cancellation schedules have been tested. Mathematical claims rest on the stated proofs and hypotheses; implementation confidence additionally rests on independent replay, adversarial examples, and the retained regression.

8 Consequences for the quasi-polynomial program

The present results make two costs more explicit. A geometric collapse epoch need not pay repeated global cohomology elimination, and a marked attachment query need not carry one additive coordinate per marked half-edge. Neither optimization supplies a bound on the number of retriangulations or useful surface choices.

The most useful form of a future global theorem would control three quantities simultaneously: the number of candidate continuations at a state, the number of successive decisions or resets, and the bit size of every state. Theorem 4.3 settles the third quantity for the specific transported-cochain move sequence, once the number of upward moves is bounded. Corollary 4.4 states the remaining coverage assumption without suppressing the choice of downward policy.

There is also an unavoidable output-size issue for a literal full-cut triangulation: arbitrarily many parallel normal components can leave arbitrarily many intervening product components. A compressed cutter must retain multiplicities, use an output-sensitive bound, or restrict its input components. The unmarked-cycle histogram and binary normal coordinates in this package are designed to preserve this information. Expanding them at the next interface would discard the gain.

8.1 A proved score refinement ready for implementation

The splitting example shows why raw Euler gain can reward the creation of vertex-link spheres. The following exact refinement removes that particular effect without restoring a global computation at every candidate. It is a proved data-structure lemma; the refined scorer is not implemented or benchmarked in this package.

For each global vertex v , let C_v be its local tetrahedral corners and let a_c be the triangle coordinate at corner c . The maximal vertex-link multiplicity is

$$m_v = \min_{c \in C_v} a_c.$$

Writing L_v for the vertex-link normal vector, define $\text{peel}(a) = a - \sum_v m_v L_v$. These subtractions preserve nonnegativity, matching, and the quadrilateral condition. In normal position the innermost complete triangle layers form the corresponding disjoint vertex-link components. With $\lambda_v = \chi(L_v)$, equal to one at a boundary vertex and two at an interior vertex,

$$\chi(\text{peel}(a)) = \chi(a) - \sum_v \lambda_v m_v.$$

This operation removes vertex links only; it does not include the quadrilateral-gcd division in the existing full canonical-core routine.

Proposition 8.1 (Thirteen corner records suffice). *During one scoring pass on a fixed triangulation and coherent vector, retain at each global vertex its thirteen least triangle-coordinate records, or all records if fewer exist. Break ties by corner identity. After this $O(t)$ -operation preparation, the exact vertex-link minima after each legal 3–2 candidate, and hence its peeled Euler jump, are computable in $O(1)$ integer operations per candidate.*

Proof. A candidate removes twelve old corners and supplies eight new corners. If any old corner of an affected global vertex survives, its least surviving record is among the first thirteen: otherwise at least thirteen records would have been removed. Remove the deleted corner identities from the

retained list and compare the least survivor with the at most eight new coordinates. If no old record survives, use infinity for that temporary minimum; the legal replacement supplies a new corner for the same global vertex.

There are at most five affected global vertex classes, even if some formal vertices are identified. A constant-capacity heap constructs each thirteen-record list in linear total work. The legal relative-boundary move preserves the global vertices and their boundary/interior types, so their link Euler characteristics are unchanged. All unaffected triangle coordinates remain fixed under cochain transport. These facts give both the claimed minima and the bound. $\square$

In particular,

$$\Delta\chi_{\text{peeled}} = 2g - \sum_{\text{affected } v} \lambda_v(m'_v - m_v). \quad (7)$$

The counterexample has raw gain four and peeled gain zero, because its two new vertex-link spheres account for the full difference. The peeled piece count has the similarly exact change

$$\Delta P_{\text{peeled}} = \Delta P_{\text{raw}} - \sum_{\text{affected } v} (m'_v |C'_v| - m_v |C_v|),$$

where $\Delta P_{\text{raw}} = -\Delta P$ in the collapse direction.

Corollary 8.2 (Peeled piece count is nonincreasing). *Under coherent 3–2 transport, the global vertex-link multiplicities satisfy $m'_v \geq m_v$, and the number of normal pieces after vertex-link peeling does not increase. This statement excludes global regauging and quadrilateral-gcd division.*

Proof. Put

$$a = [\min(U, x) - L]_+, \quad b = [U - \max(L, z)]_+.$$

These are the minimum old triangle coordinates at the lower and upper formal apex. The new apical minima are $a + g$ and $b + g$. At a belt vertex, its minimum over the local corners is unchanged. Indeed, the triangle coordinate at height h is its distance to the interval spanned by the other three tetrahedral heights. The two old intervals overlap on the apex interval; the two new intervals overlap on the interval of the other belt vertices. Both unions are the interval spanned by the other four heights. Taking the minimum distance gives the same value.

Taking further minima over identified formal vertices and over unchanged exterior corners proves $m'_v \geq m_v$. Let r_v count the formal apices identified with v . Then $|C'_v| = |C_v| - 2r_v$ and $\sum_v r_v m_v \leq a + b$, also when the apices are identified. Projection onto $[L, U]$ gives

$$a = \text{clamp}_{[L, U]}(x) - L, \quad b = U - \text{clamp}_{[L, U]}(z).$$

Since $x \leq z$, we have $a + b \leq U - L \leq \text{rng}(L, U, y)$, and hence $\Delta P = \text{rng}(L, U, y) + a + b \geq 2(a + b)$. Writing $\delta_v = m'_v - m_v \geq 0$, the peeled change is

$$\begin{aligned} \Delta P_{\text{peeled}} &= -\Delta P + 2 \sum_v r_v m_v - \sum_v \delta_v |C'_v| \\ &\leq -2(a + b) + 2(a + b) = 0. \end{aligned}$$

$\square$

This corollary is independently checked on all 3,125 small-height assignments and all 52 partitions of the five formal vertices, with additional exterior-minimum controls: 812,500 exact algebraic cases

in total. These tests do not assert that every partition is realized by a legal ambient triangulation. The corollary does not establish monotonicity of the peeled Euler characteristic, or a bound on the number of choices needed to discover an essential disc.

These summaries must be rebuilt after committing a move or maintained by a separately justified dynamic structure. Repeatedly deleting from a thirteen-element list without replenishing it is invalid. The lemma also requires that coordinates outside the candidate region remain unchanged; it does not cover simultaneous global regauging. Whether the resulting priority improves essential-disc discovery is a separate experimental question, and non-vertex-link spheres still require component analysis.

8.2 Proposed research questions and concrete milestones

- Q1. A bounded-escape coverage theorem.** Does a useful, explicitly defined class of unknot diagrams have a successful coherent-fibre witness in the deterministic macrofamily of Corollary 4.4, or its explicit upward-burst extension, at depth $O(\log n)$? A first target should name the diagram class and the exact downward policy. A counterexample should distinguish lack of a short macro-path from mere failure of one bounded execution.
- Q2. Exchange of downward moves.** Under what conditions do two eligible collapses commute, or admit a common descendant with the same coherent boundary data? Proving a local exchange theorem could justify collapsing many orders to one macrostate. Without such a theorem, retaining one greedy order can lose coverage.
- Q3. Incremental geometric preparation.** Maintain signed edge classes, degree-three candidate sites, and the five-height scores across a move. The current implementation rebuilds prepared geometry. An update algorithm must handle edge classes affected by global face identifications, not only the visible formal vertices, and must have an independently replayable update contract.
- Q4. Compact transcripts with local reconstruction.** The supplied transcript stores whole intermediate triangulations and can occupy quadratic space. Can an independent consumer reconstruct each next triangulation from the local proof and a stable cell-name map, retaining only a linear-size current state and a nearly linear transcript?
- Q5. Gauge optimization after transport.** Study reuse of the minimum-span dual certificate and the Euler-optimal face after a local move. A correct incremental method must account for the changed tetrahedral objective and distinguish harmless local height normalization from a global vertex coboundary. Compare both candidate coverage and full replay cost with a fresh exact solve.
- Q6. Peeled scoring and connectivity through the exceptional slab.** Implement Proposition 8.1 and compare the refined score with raw Euler gain on source-bound examples. Give a precise global description of the nested annulus-to-disc replacements, including which components split and whether a selected essential-boundary component can be followed without a fresh full census. The additive Euler formula alone is insufficient for this purpose.
- Q7. A three-dimensional attachment compiler.** From a normal surface and a triangulation, construct the nonparallel cut pieces and the degree-two curve systems along each vertical parallelity component. Feed their marked occurrences to the present ordering interface. The milestone is one independently checked cut with ordered attaching squares and horizontal-side correspondence.

- Q8. Signed endpoint transport.** Implement the two-sheeted cover extension for side parity and compare it with direct signed path weights. Specify what happens to an unmarked orientation-reversing circle and ensure that the cover does not erase bundle multiplicity.
- Q9. Higher bounded endpoint valence.** For an interface whose residual components have at most d endpoints, can the first d exact power sums replace a unit-vector weight system with a practical decoder? Newton identities give an algebraic route, but root recovery, multiplicity, integer coefficient growth, and independent replay all need explicit bounds. The two-endpoint case here is unusually simple and should remain the baseline for performance comparisons.
- Q10. Certified input-sensitive scheduling.** Predict when moment boundary queries or cochain descent will save actual time. The scheduler must be evaluated on complete recognition and should charge failed candidate searches. A local kernel speedup does not justify unconditional activation on every diagram.
- Q11. An adversarial geometric corpus.** Build reproducible unknotted families with growing upward escape requirements, and nontrivial knots whose low-complexity coherent candidates are misleading. Retain all negative results and resource-limited runs. Include true source diagrams, not only triangulations with a known construction history.
- Q12. Formal verification of the small kernels.** The five-height identities, endpoint-moment decoder, degree-two deletion lemma, and source-consumer implication are suitable small formalization targets. Their machine checking would reduce the trusted mathematical core while leaving the larger AHT and three-manifold infrastructure as explicit dependencies.

A Implementation and reproduction guide

A.1 Archive layout and integration

The archive contains a complete source snapshot needed by the supplied tests, an additive patch against the audited commit, the article sources and rendered PDF, raw measurements, certificate-bearing audit records, compact source cases, reproduction scripts, and provenance notes. The `fast/` snapshot preserves the pinned files and adds the new modules. Its sibling `reports/` directory contains only historical oracle dependencies required by the existing tests.

The patch uses paths rooted at `Topology/UnknotRecognition/fast/`. It adds the feature modules, tests, and research drivers. It does not alter the production recognition schedule. The new article and the retained data are separate reviewable artifacts; the repository's intake process can place them under its next report number.

From a checkout at the pinned revision, inspect and check the patch:

```
git apply --stat /path/to/integration.patch
git apply --check /path/to/integration.patch
```

The package build verifies that the patch applies to the pinned source and reproduces its included added files. A later repository revision should be checked independently for path conflicts before applying it.

A.2 Python setup and tests

The new native algorithms use the Python standard library. The maintained project supports Python 3.10 or later. This research environment uses Python 3.12.14; use Python 3.12 for the pinned optional package set. Regina is optional for most native tests but required to repeat the advertised independent Regina comparisons. The archive gives the exact optional package versions in `requirements-research.txt`; it does not bundle third-party binary distributions.

A reproducible environment can be prepared at the extracted archive root:

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements-research.txt
python reproduce/run_checks.py --full
```

Without the optional dependencies, native feature checks can be run with:

```
python reproduce/run_checks.py
```

The latter command uses standard-library test discovery and verifies retained native proofs. Its output identifies optional comparisons that were skipped. For the explicit repository regression:

```
cd fast
python -m unittest discover -s tests -v
```

A.3 Minimal source-bound example

From the supplied `fast/` directory:

```
from fastunknot import Diagram
from fastunknot.normal_transport import transport_seed_decide
from fastunknot.normal_transport_verify import (
    verify_transport_disk_certificate,
)

diagram = Diagram.from_braid(2, [1])
answer = transport_seed_decide(
    diagram, strategy="score", max_work=2_000_000
)
assert answer["status"] == "UNKNOT"
assert verify_transport_disk_certificate(
    diagram, answer["certificate"]
)
```

The function accepts optional boundary shellings and initial minimum-span optimization. It performs no subsequent cohomology solve after a move. A failed or resource-limited search returns `INCONCLUSIVE`. The certificate schema is `diagram-transport-disc-v1`; it includes the source PD, canonical exterior, optional shelling transcript, initial heights, every triangulation and cochain-transport proof, final coordinates, and a positive essential-disc component certificate.

No rank-one or primitivity claim about the initial heights is needed by the positive consumer: the independently established essential disc in the verified exterior is sufficient. The producer uses a primitive class to make its search useful. This separation keeps the consumer's mathematical responsibilities smaller.

A.4 Low-level geometric and boundary APIs

```
from fastunknot.cocycle_transport import (
    bipyramid_cocycle_score,
    cocycle_collapse_candidates,
    descend_cocycle,
)
score = bipyramid_cocycle_score([0, -1, 1, 3, 4])
assert score["euler_loss"] == 4
assert score["normal_disc_increase"] == 5
```

The five-height input order is belt first, then apices. A transport proof uses schema `pachner-cocycle-transport-v1`. The mathematical formulas are valid for arbitrary integral heights; the native geometry wrapper has the compact orientable one-torus-boundary contract stated in Section 2.

```
from normal_orbit_research.fixtures import layered_torus
from fastunknot.marked_boundary import normal_marked_boundary_order
from fastunknot.marked_boundary_verify import (
    verify_normal_marked_boundary_certificate,
)

triangulation, coordinates = layered_torus(8)
marks = [0]
answer = normal_marked_boundary_order(
    triangulation, coordinates, marks,
    weight_encoding="moments", record_certificate=True
)
assert answer["status"] == "COMPLETE"
assert verify_normal_marked_boundary_certificate(
    triangulation, coordinates, marks, answer["certificate"]
)
```

The lower-level `marked_boundary_order(size, pairings, marks, ...)` accepts a validated raw degree-two multigraph encoding. The reference `weight_encoding="one_hot"` is retained. The default moment mode changes the query encoding, not the returned cyclic solution. An optional `start_half_edge` fixes the direction of one marked component. Unmarked circles remain present in the output histogram.

A.5 Recreate audits and benchmarks

From `fast/`, the following commands write new records into `../reproduced/`; they preserve the supplied measurements:

```
python -m transport_research.audit local \
    --output ../reproduced/cocycle-local.json
python -m transport_research.audit corpus \
    --corpus ../reproduce/cocycle-source-cases.json \
    --output ../reproduced/cocycle-corpus.json
python -m transport_research.audit benchmark \
    --sizes 16 32 64 128 256 --rounds 5 \
    --output ../reproduced/cocycle-benchmark.json
python -m transport_research.splitting \
    --output ../reproduced/cocycle-splitting.json
```

```
python -m marked_boundary_research.matched_final \
  --output ../reproduced/marked-boundary-paired.json
```

The local transport audit uses Regina unless `-no-regina` is specified. The final paired boundary driver records individual samples and exact equality checks. The broader native-boundary experiment can also be repeated with:

```
python -m marked_boundary_research.benchmark \
  --output ../reproduced/marked-boundary.json
```

Timing benchmarks should run sequentially with other heavy jobs paused. Exact numerical outputs, counts, and accepted certificates are reproducibility targets; machine-dependent elapsed times are not expected to match bit for bit.

A.6 Build the article

The included table and figure sources reproduce the paper’s displays from the retained JSON measurements. The already generated figures permit a PDF rebuild without Python plotting packages. From `article/`:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  unknot_geometric_transport.tex
```

Rebuilding the tables and plots first uses:

```
python reproduce/build_tables.py
```

from the archive root; the plotting dependency is listed separately in the research environment file. The final PDF was rendered to page images and visually inspected for broken equations, table overflow, and pagination.

B Claim ledger

Claim	Evidence	Scope
Euler and disc jumps	Relative cell proof, independent coarea table, exhaustive small-height checks	Reselected coherent fibre of a transported cocycle; total Euler characteristic
Class preservation	Five-height lift and checked relative-boundary move	Integral cohomology class, not fibre isotopy
Polynomial downward epoch	Tetrahedron decrease and invariant edge max-norm	One fixed downward policy, no upward escape coverage
Conditional quasi-polynomial macrosearch	Explicit state-tree count and coordinate-bit bound	Requires a witness in the specified bounded-depth macrofamily; not shipped exhaustive search
Four-coordinate endpoint decoding	Degree-two deletion lemma and exact quadratic identity	At most two distinct endpoint occurrences per residual path; not arbitrary multisets
Marked cyclic order	Independent residual graph and orbit replay	Distinct explicit marks, raw edge occurrences, exact gaps
Peeled piece monotonicity	Local triangle minima and exact corner counts; 812,500 algebraic checks	Coherent collapse, vertex-link peeling only; no regauging or gcd division
Thirteen-record scoring refinement	Constant deletion count and least-surviving-record proof	One fixed scoring pass; proposed scorer not implemented
Diagram-bound unknot witness	Exterior, shelling, transport, and positive normal-disc checks	Supplied positive transcript; a failed search is inconclusive
Measured speed improvements	Paired retained samples and independent output comparisons	Workloads and units identified in the experiment tables
General quasi-polynomial recognition	Not proved by this package	Global search and full cut-state construction remain open implementation requirements

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